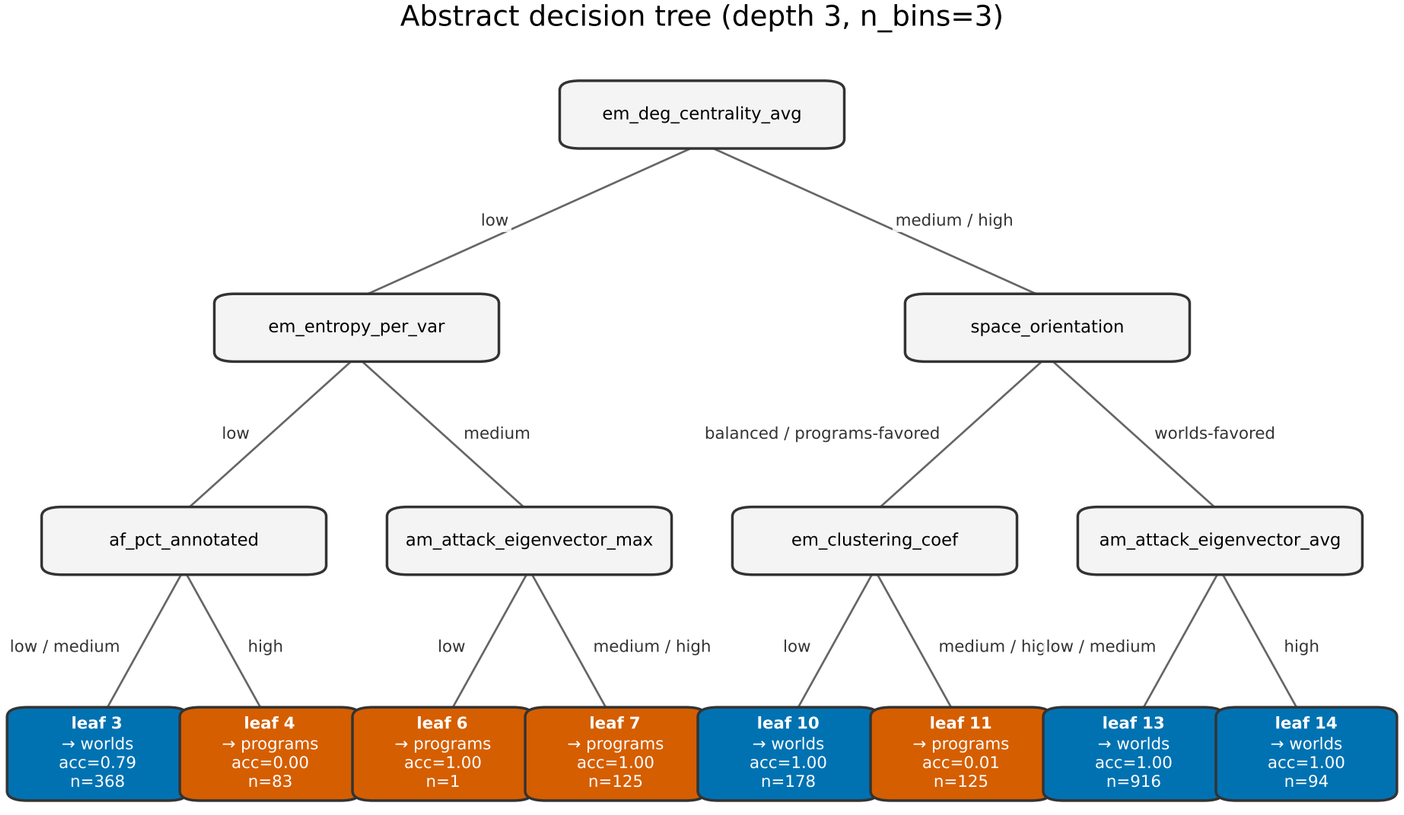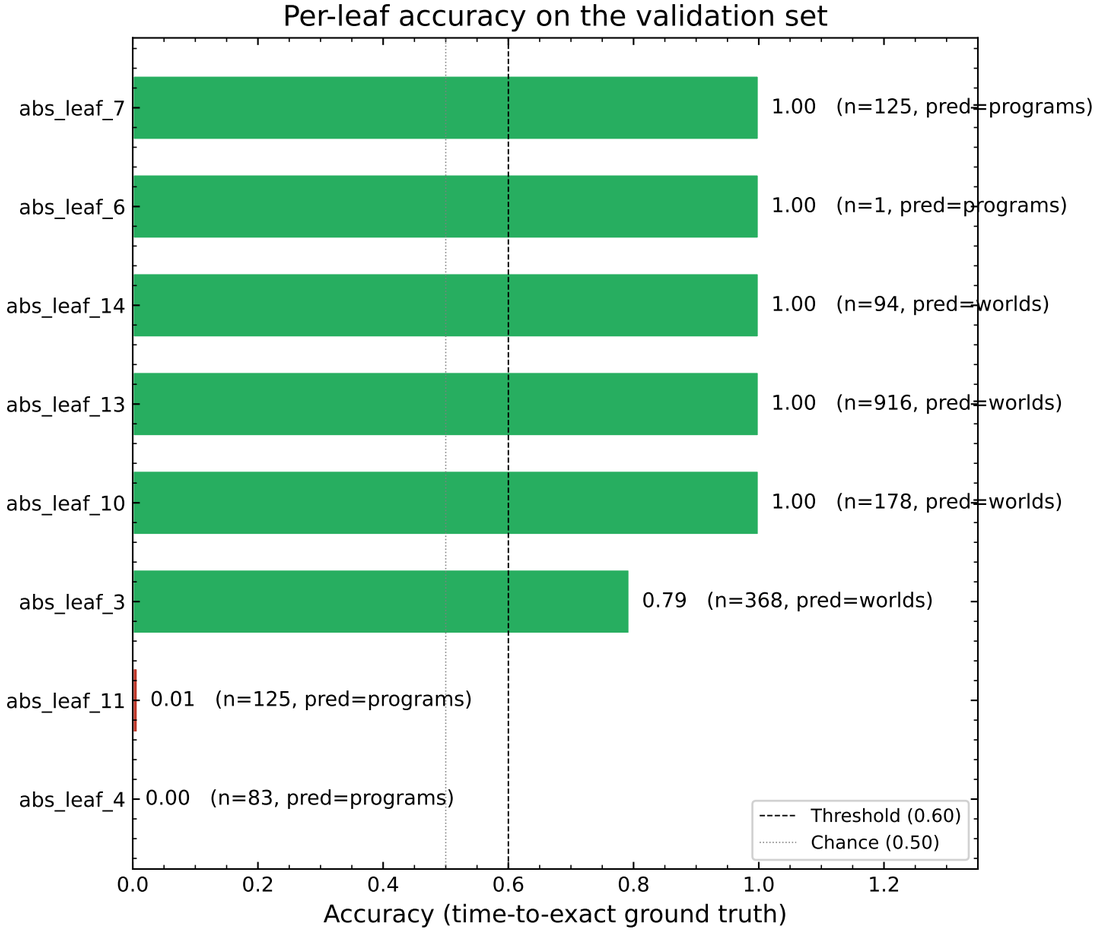Abstract tree structure and its validation on the time-to-exact ground truth (overall accuracy = 0.850)